DRUGS · NASAL PREPARATIONS

Saline irrigation

Saline irrigation washes mucus, crusts, allergens and inflammatory mediators from the nose, improves ciliary function and delivers any added drug to the mucosa. High-volume, low-pressure isotonic or buffered saline is recommended first-line in chronic rhinosinusitis and after sinus surgery, and is useful in allergic rhinitis and atrophic rhinitis. It is cheap and safe if the water is clean.

1 CHAPTER ONE At a glance

The treatment on one page.

1.1 Summary

SALINE IRRIGATION

	SUMMARY
Forms	High-volume (about 240 ml) low-pressure squeeze bottle or neti pot; low-volume sprays and drops; isotonic (0.9%) or hypertonic (2% to 3%); buffered with sodium bicarbonate
How it works	Mechanical clearance of mucus, crusts and allergens; improved mucociliary clearance; dilution of inflammatory mediators
ENT uses	Chronic rhinosinusitis (first-line, with nasal steroid); after endoscopic sinus surgery; allergic rhinitis (adjunct); atrophic rhinitis and crusting; granulomatosis with polyangiitis; after radiotherapy to the nasal cavity
Dose	150 to 240 ml into each nostril once or twice daily (high-volume); after surgery more often, starting the day after the operation
Safety	Use boiled and cooled, distilled or sterile water; clean and dry the bottle; replace it every few months

2 CHAPTER TWO Using it in ENT

Evidence and practice.

2.1 Evidence

High-volume saline irrigation improves symptoms in chronic rhinosinusitis more than low-volume sprays and is recommended by EPOS and NICE. After sinus surgery it removes crusts and clot and reduces adhesions. In children with allergic rhinitis, saline reduces symptoms and medication use.

2.2 Isotonic or hypertonic

Hypertonic saline may improve ciliary beat and decongest a little more, but stings more; isotonic or buffered isotonic saline is usually better tolerated.

2.3 Technique

Lean forward over the sink, mouth open, head turned slightly, squeeze gently into the upper nostril so the solution runs round the septum and out of the other side, then blow gently. Use the solution at body temperature.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- Stinging, ear fullness and fluid running from the nose later.
- Contaminated equipment can carry bacteria; **tap water** used without boiling has rarely caused fatal amoebic meningoencephalitis (Naegleria fowleri) and Acanthamoeba infection.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- High-volume versus low-volume irrigation.
- Water safety.
- Its place as first-line therapy in chronic rhinosinusitis.

4.2 Common pitfalls

COMMON PITFALLS

- Unboiled tap water.
- Prescribing a low-volume spray where a high-volume rinse is needed.

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